

Nagendra Chowdary

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PROFESSIONAL SUMMARY:

- AI/ML Engineer with **11+ years of experience** designing, developing, and deploying machine learning, deep learning, NLP, GenAI, RAG, and predictive analytics solutions across healthcare, banking, financial services, retail, supply chain, and telecom domains.
- Strong experience building **GenAI and RAG-based applications** using LangChain, LangGraph, LlamaIndex, Amazon Bedrock OpenAI APIs, Hugging Face Transformers, FAISS, Chroma, Pinecone, Redis Vector Cache, and OpenSearch Vector Search.
- Experienced in developing **end-to-end AI/ML pipelines** covering requirement analysis, data extraction, data profiling, feature engineering, model development, validation, deployment, monitoring, and production support.
- Skilled in building **AWS-based AI/ML solutions** using S3, EC2, Lambda, Glue, SageMaker, EMR, Redshift, Athena, Kinesis, CloudWatch, CloudTrail, ECR, ECS/EKS, IAM, and API Gateway.
- Developed production-ready machine learning models for **fraud detection, AML transaction monitoring, risk scoring, churn prediction, demand forecasting, inventory optimization, member risk scoring and customer segmentation**.
- Hands-on experience with **NLP and transformer-based models** including BERT, DistilBERT, ClinicalBERT, BioBERT, spaCy, Hugging Face Transformers, TF-IDF, Word2Vec, text classification, entity extraction, summarization, and semantic similarity.
- Built scalable data processing and feature engineering pipelines using **Python, SQL, PySpark, Spark SQL, Pandas, NumPy, Airflow, AWS Glue, Athena, Redshift, Snowflake, and relational databases**.
- Experienced in developing and exposing AI/ML capabilities through **FastAPI, Flask, REST APIs, Docker, Kubernetes, ECS/EKS, SageMaker endpoints, and CI/CD pipelines using GitHub Actions, GitLab CI/CD, and Jenkins**.
- Strong MLOps and LLMOps background with **MLflow, SageMaker Pipelines, LangSmith, Langfuse, model versioning, experiment tracking, drift monitoring, prompt evaluation, response quality tracking, and production observability**.
- Applied model explainability, governance, and compliance practices using **SHAP, LIME, Responsible AI monitoring, IAM, RBAC, encryption, KMS, PHI/PII masking, audit logging, HIPAA-aware data handling, and financial model risk controls**.
- Built dashboards and analytical reporting solutions using **Tableau, Power BI, SQL, Python visualizations, CloudWatch logs, and KPI reporting** to track model performance, fraud trends, churn risk, inventory KPIs, care operations and GenAI adoption.
- Proven ability to work with cross-functional teams including product owners, data engineers, cloud engineers, compliance teams, business stakeholders, fraud analysts, care operations teams, and marketing users to deliver scalable AI/ML solutions with measurable business impact.

TECHNICAL SKILLS:

- **Programming & Data Science:** R / Python 3.x / PySpark
- **Programming Languages:** Python, R, C, C++, Java, SQL, PL/SQL, Bash/Shell, Scala, TypeScript (basic)
- **Databases:** Microsoft SQL Server, MySQL, Oracle 12c, DB2, Teradata, Cassandra, MongoDB, PostgreSQL, Redis, Snowflake, Amazon Redshift
- **Big Data & Processing:** Hadoop, HDFS, HBase, Hive, Apache Spark (Core, Streaming, MLlib), Kafka, Kafka Streams, Airflow, Sqoop, Oozie
- **Data Visualization:** Tableau, Power BI, Looker, Matplotlib, Seaborn, Plotly
- **Artificial Intelligence:** NLP, Computer Vision, Speech Recognition/Generation, Conversational AI, GANs, Expert Systems, Hugging Face Transformers, OpenAI API
- **Machine Learning & Deep Learning:** Classification, Regression, Clustering, Dimensionality Reduction, Semi-/Self-Supervised Learning, CNNs, RNNs, Transformers, Reinforcement Learning, Transfer Learning, Ensemble Methods, TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, XGBoost, LightGBM, CatBoost, Optuna (Hyperparameter Tuning), MLflow (Model Tracking), DVC (Data Version Control)
- **LLM & GenAI Frameworks:** LangChain, LlamaIndex, Autogen AI, Semantic Kernel, PromptFlow, Gradio, Claude, AWS Bedrock, Gemini API, CrewAI, OpenAI Assistants API, Azure AI Agent Service, Vertex AI Agent Builder

- **Vector Databases & Semantic Search:** FAISS, Pinecone, ChromaDB, Redis (Vector Cache), Elasticsearch (Hybrid Search)
- **Data Lakehouse:** Delta Lake, Apache Iceberg
- **Stream Processing:** Apache NiFi, Apache Beam, Apache Pulsar
- **Data Science Techniques:** EDA, Feature Engineering, Time Series Analysis, Anomaly Detection, Recommender Systems, Text/Image Generation, Text-to-Image, RLHF, ChatGPT
- **MLOps & Model Deployment:** Docker, Kubernetes, Helm, Flask, Django, FastAPI, TensorFlow Serving, Seldon Core, REST API Development, Nest.js (intro)
- **DevOps & CI/CD:** Git, GitHub Actions, GitLab CI/CD, Bitbucket Pipelines, Jenkins, CircleCI, Terraform, Ansible
- **Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker, EMR, Redshift, CloudWatch), Azure (ML Studio, Data Factory, Synapse), GCP (BigQuery, Vertex AI, Dataflow, AutoML)
- **Version Control & Collaboration:** Git, GitHub, GitLab, Bitbucket, Jira, Confluence
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack, AWS CloudWatch
- **Security & Data Governance:** Role-Based Access Control (RBAC), Data Encryption, IAM, Data Masking, GDPR/CCPA Compliance, Key Management Service (KMS)

EXPERIENCE:

Client: Discover

Role: Senior AI/ML Engineer

Mar 2025 – Present | Remote

Project: AI-Driven Fraud Intelligence, Financial Forecasting, RAG Assistant & Agentic AI Platform

- Designed and delivered **AI-driven financial forecasting models** using **Auto-Sklearn, H2O.ai, Python, Spark ML, and statistical modeling techniques** to support market trend prediction, portfolio risk estimation, and business planning, improving forecast accuracy by **15%**.
- Built scalable **fraud detection pipelines** using **AWS SageMaker, PySpark, Kinesis Data Streams, Lambda, and real-time feature processing**, improving true-positive fraud detection rates by **12%** and reducing manual fraud case reviews by **20%**.
- Engineered advanced fraud-risk features including **transaction velocity thresholds, geolocation mismatches, device fingerprinting, merchant behavior, customer activity patterns, and anomaly signals**, reducing false-positive alerts by **18%**.
- Developed **AML transaction monitoring models** for segmentation, suspicious activity detection, SAR alert prioritization, and compliance review using **AWS Glue, Athena, PySpark, SQL, and feature engineering pipelines**, reducing SAR case processing time by **30%**.
- Partnered with compliance, model governance, fraud operations, and risk stakeholders to align AI/ML workflows with **Model Risk Management standards**, integrating **SHAP, LIME, traceability, audit logging, and explainability reports** into model review processes.
- Delivered advisor-facing **LLM and RAG-based chatbot solutions** using **GPT-4, LangChain, Bedrock/OpenAI APIs, vector search, prompt engineering, and FastAPI**, reducing average advisor response time by **40%**.
- Built full-stack conversational AI services using **Python, FastAPI async APIs, REST services, Docker, Kubernetes, and internal advisor platforms**, enabling secure integration of AI search, summarization, and recommendation capabilities into business applications.
- Designed and prototyped **multi-agent orchestration workflows** using **LangGraph, AutoGen AI, Langfuse, Bedrock/OpenAI APIs, tool calling, memory/state handling, and evaluation traces** for fraud triaging, decision support, automated RCA, and task routing.
- Implemented **supervisor and plan-execute agent patterns** in **LangGraph** with planner, retriever, validator, escalation, and action nodes to support controlled agentic workflows for fraud investigation and alert review.
- Optimized enterprise **RAG pipelines** using advanced chunking, metadata tagging, hybrid retrieval, semantic reranking, knowledge graph augmentation, Redis vector caching, Pinecone, and PostgreSQL JSONB, improving retrieval precision and reducing hallucination risk.
- Automated **Jira ticket creation, status updates, RCA summaries, and workflow recommendations** through AI agents integrated with internal tools, reducing mean time to resolution by **28%** and improving operational handoff consistency.

- Designed and deployed **Vertex AI pipelines** for model training, evaluation, batch scoring, and real-time serving, integrating with **BigQuery ML** and **GCP Dataflow** to support large-scale analytics and streaming ingestion.
- Improved LLM performance and cost efficiency through **prompt optimization, response caching, token management, LoRA/PEFT experiments, LangSmith evaluation, and prompt regression testing**, improving response quality while reducing compute overhead.
- Automated AI/ML infrastructure provisioning using **Terraform, Ansible, Cloud Composer, SageMaker Pipelines, Step Functions, and CI/CD workflows**, reducing environment setup time by **40%** and enabling faster rollback and drift detection.
- Deployed real-time inference using **AWS SageMaker, EC2, DynamoDB, Lambda, Kinesis, Docker, and AWS EKS**, supporting high-availability model serving with **99.9% uptime** and inference latency under **200ms**.
- Built and evaluated **classification, anomaly detection, boosted tree, neural network, GAN-based synthetic data, and multilingual NLP pipelines** using **XGBoost, LightGBM, TensorFlow/PyTorch, BERT, spaCy, and Hugging Face Transformers** for fraud scoring, risk profiling, summarization, NER, and market signal detection.
- Configured observability and Responsible AI monitoring using **Cloud Monitoring, Cloud Logging, ELK, WhyLabs, LangSmith, Langfuse, CloudWatch, and custom dashboards** to track model drift, LLM quality, throughput, latency, token usage, anomaly patterns, and production reliability.
- Managed secure delivery of AI/ML solutions using **GitLab CI/CD, Docker, Kubernetes, AWS EKS, IAM roles, VPC controls, KMS encryption, Jira, and Confluence**, ensuring reproducible builds, Agile delivery, and compliance with financial data security requirements.

Client: Belong Health

Role: AI/ML Engineer

Aug 2022 – Feb 2025 | Radnor, PA

Project: AWS-Based Healthcare GenAI Platform, RAG Knowledge Assistant & Clinical NLP Automation

- Led the design and development of **GenAI, RAG, NLP, and ML solutions on AWS** for healthcare operations, care management, member engagement, clinical document intelligence, and internal decision-support use cases.
- Built and enhanced **AWS-based AI/ML solutions** using **Python, PySpark, AWS S3, Glue, Lambda, Redshift, Athena, SageMaker, and OpenSearch** to process healthcare claims, eligibility, care gap, and document data.
- Developed predictive ML models for **member risk scoring, care gap prioritization, outreach targeting, claims trend analysis, and workload forecasting** using **Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, and PySpark**.
- Created feature engineering pipelines from structured healthcare data, including **claim frequency, utilization history, care gap status, category indicators, prior authorization patterns, member engagement history**.
- Built healthcare NLP workflows using **BERT, ClinicalBERT Hugging Face Transformers, spaCy, and TF-IDF** for clinical note classification, entity extraction, keyword extraction, intent detection, and support ticket categorization.
- Developed clinical note summarization and case review support workflows using **LLM APIs and transformer-based NLP models** to summarize long notes, extract follow-up actions, and reduce manual review effort by **25%**.
- Designed and developed a **RAG-based healthcare knowledge assistant** to help internal users search SOPs, policy documents, provider manuals, clinical guidelines, FAQs, and care management documents.
- Built document ingestion pipelines using **Amazon S3, AWS Lambda, AWS Glue, Amazon Textract, Python, and PySpark** to process PDFs and healthcare documents through text extraction, cleaning, chunking, metadata tagging, embedding generation, and vector indexing.
- Implemented semantic search and retrieval workflows using **LangChain, LlamaIndex, FAISS, Chroma, Amazon OpenSearch Vector Search, embedding models, metadata filters, hybrid search, reranking, and prompt templates**, improving document search relevance by approximately **30%**.
- Developed **LangGraph-based controlled agent workflows** with retriever, summarizer, validator, guardrail, and escalation nodes to support member inquiry lookup, care gap review, policy search, case summarization, and human-in-the-loop.
- Built backend APIs using **Python, FastAPI, Flask, REST APIs, Amazon API Gateway, AWS Lambda, SQL, and containerized services on ECS/EKS** to expose AI/ML, RAG search, summarization, and classification services to internal applications.
- Implemented GenAI safety and healthcare compliance such as **PHI/PII masking, IAM-based access, role-based permissions, prompt validation, response filtering, source citations, encryption, and HIPAA-aware data handling**.
- Used **MLflow and Amazon SageMaker** for experiment tracking, model versioning, metric comparison, artifact logging, and reproducible model development across ML, NLP, and GenAI workflows.

- Supported deployment and production readiness using **Docker, GitHub Actions/Jenkins, Amazon ECR, ECS/EKS, CloudWatch, CloudTrail, and AWS CI/CD workflows**, reducing deployment issues by approximately **20%**.
- Built monitoring dashboards and reports using **Tableau, SQL, CloudWatch logs, and Python visualizations** to track model performance, retrieval quality, API latency, failed queries, token usage, user feedback, and care operations KPIs.

Client: Bank of America

Role: Machine Learning Engineer

Dec 2019 – July 2022 | Plano, TX

Project: Fraud Risk Scoring, Transaction Intelligence & NLP-Based Case Analytics Platform

- Worked on an enterprise **fraud risk scoring and transaction intelligence platform** supporting fraud detection, suspicious activity identification, transaction monitoring, customer dispute classification.
- Owned the end-to-end **ML lifecycle**, including business requirement analysis, data preprocessing, feature engineering, model development, validation, deployment support, monitoring, and production enhancement.
- Processed large-scale banking datasets from **transaction systems, customer profiles, account activity, merchant data, device logs, login history, digital banking events, dispute records, fraud case notes, analyst comments, and enterprise data warehouses**.
- Built data preparation pipelines using **Python, SQL, PySpark, Spark SQL, Pandas, and NumPy** to clean, join, transform, and standardize large-scale financial datasets for machine learning model development.
- Engineered fraud-risk features such as **transaction frequency, amount deviation, merchant category risk, login velocity, device behavior, geographic mismatch, failed transaction attempts, account age, customer behavior changes, and historical fraud indicators**.
- Developed fraud detection, anomaly detection, risk scoring, and suspicious activity classification models using **Logistic Regression, Random Forest, Gradient Boosting, XGBoost, Isolation Forest, K-Means, and NN**.
- Handled fraud class imbalance using **SMOTE, class weights, threshold tuning, precision-recall optimization, cross-validation, and false-positive reduction strategies**, improving fraud detection efficiency while reducing unnecessary analyst workload.
- Built NLP pipelines to process unstructured banking text, including **fraud case notes, customer dispute narratives, support tickets, complaint descriptions, and compliance investigation notes**.
- Fine-tuned **BERT and DistilBERT transformer models** using **Hugging Face Transformers and Pytorch** for fraud reason categorization, dispute type prediction suspicious activity classification, and case prioritization.
- Combined structured transaction features with NLP-based text embeddings from **BERT, DistilBERT, TF-IDF, and Word2Vec** to improve fraud risk scoring, case triaging, and analyst review prioritization.
- Evaluated ML and NLP models using **precision, recall, F1-score, ROC-AUC, PR-AUC, confusion matrix, false-positive rate, false-negative rate, KS score, lift charts, accuracy, and actual vs. predicted analysis**.
- Used **MLflow** to track experiments, model versions, parameters, metrics, artifacts, feature sets, and model comparison results across development, validation, and deployment preparation cycles.
- Built batch scoring workflows using **PySpark, Python, SQL, Airflow, and enterprise scheduling tools** to generate fraud risk scores, alert outputs, and analytical datasets for downstream fraud operations teams.
- Supported model deployment and production readiness using **AWS S3, EC2, EMR, SageMaker, Docker, Kubernetes, Flask/FastAPI APIs, and internal banking cloud environments**.
- Created model serving components to expose fraud scores, NLP classification outputs, and risk indicators to internal applications, dashboards, and review workflows.
- Collaborated with DevOps and engineering teams to containerize ML components, support **CI/CD workflows**, and promote models across **development, testing, UAT, and production** environments.
- Built monitoring workflows and dashboards to track **model drift, feature stability, score distribution changes, prediction volume, data quality issues, false positives, false negatives, and model performance degradation** after production release.
- Developed dashboards and analytical reports using **Power BI, SQL, and Python visualizations** to help fraud analysts and stakeholders review flagged transactions, risk patterns, alert volumes, model performance, NLP classification results, and operational KPIs.
- Partnered with fraud operations, compliance, model validation, data engineering, and application teams to document **model assumptions, feature logic, validation results, monitoring approach, data lineage, and production support details**.

- Improved fraud review efficiency by refining model thresholds, incorporating analyst feedback, validating predictions against historical fraud cases, and enhancing risk signals used for case prioritization and investigation support.

Client: Advance Auto Parts

Role: Senior Data Scientist

Sep 2017 – Nov 2019 | Raleigh, NC

Project: Demand Forecasting, Inventory Optimization & Retail Analytics Platform

- Worked on a large-scale **Demand Forecasting and Inventory Optimization project** to improve SKU-level, store-level, category-level, and distribution-center inventory planning across retail operations.
- Collaborated with **merchandising, supply chain, pricing, replenishment, inventory planning, and business operations teams** to understand stockout issues, excess inventory, forecast gaps, promotion impact, and replenishment challenges.
- Gathered and analyzed historical **sales, inventory, product, promotion, supplier lead-time, store, and distribution center data** from **SQL Server, Oracle, Teradata, AWS Redshift, AWS S3, and enterprise data warehouse systems**.
- Built data extraction and preprocessing workflows using **SQL, Python, Pandas, NumPy, and AWS S3 file processing** to clean, standardize, and prepare high-volume retail datasets for forecasting and ML models.
- Performed data profiling and validation to identify **missing values, duplicate records, SKU mapping gaps, abnormal sales spikes, inconsistent products, inventory mismatches, and supplier lead-time anomalies**.
- Conducted exploratory data analysis using **Python, SQL, Power BI, Matplotlib, and Seaborn** to uncover demand seasonality, sales trends, promotional lift, slow-moving SKUs, and stockout behavior.
- Designed advanced feature engineering logic including **rolling sales averages, lag variables, sales velocity, days of supply, inventory turnover, reorder frequency, promotion flags, discount indicators, supplier lead-time variance, holiday flags, and regional demand signals**.
- Developed demand forecasting models using **time-series forecasting, Linear Regression, Random Forest, Gradient Boosting, and XGBoost** to predict future sales demand at **SKU, store, category, and regional levels**.
- Created **stockout and overstock prediction models** using **Logistic Regression, Decision Trees, Random Forest, and XGBoost** to identify inventory risk based on sales velocity, on-hand quantity, replenishment frequency, supplier lead times, and historical demand patterns.
- Built customer and product segmentation models using **K-Means clustering and RFM-style analysis** to group customers and products by purchase behavior, sales frequency, category preference and movement patterns.
- Evaluated model performance using **MAPE, RMSE, MAE, accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix and actual vs. predicted comparisons** to select reliable models for business use cases.
- Used **AWS EC2-based Jupyter Notebook environments** for Python model development, feature engineering, experimentation, batch processing, and analysis of high-volume sales and inventory datasets.
- Created dashboards and executive reports using **Power BI, SQL and Python visualizations** to monitor forecast accuracy, inventory KPIs, stockout risk, overstock exposure, sales performance, and promotion effectiveness.
- Automated recurring model input preparation, data refresh, validation checks, and weekly/monthly reporting using **Python scripts, SQL queries, AWS S3 data files, and scheduled workflows**, reducing manual reporting effort and improving consistency.
- Presented model insights and business recommendations to supply chain and merchandising stakeholders, helping improve replenishment planning, reduce stockout risk, identify slow-moving inventory, and support data-driven pricing and promotion decisions.
- Supported measurable inventory improvements by helping business teams improve forecast visibility, prioritize high-risk SKUs, reduce excess inventory exposure, and make faster replenishment decisions across stores and distribution centers.

GlobalLogic

Role: Data Scientist

May 2015 – Aug 2017 | India

Project: Telecom Customer Segmentation, Churn Prediction & Retention Analytics

- Worked on an end-to-end **customer segmentation and churn analytics project** for a telecom client, covering business requirement gathering, data extraction, exploratory analysis, feature engineering, model development, validation, dashboarding, and stakeholder delivery.

- Extracted and analyzed **5M+ subscriber records** from relational databases using **SQL queries, stored procedures, joins, aggregations, and data profiling techniques** across usage logs, billing history, recharge activity, service interactions, and customer plan data.
- Performed exploratory data analysis using **Python, Pandas, NumPy, Matplotlib, Seaborn, Excel, and SQL** to identify churn patterns, customer usage behavior, recharge trends, service issues, data quality gaps, and key retention drivers.
- Identified that subscribers with **2+ consecutive months of declining data usage** were nearly **3x more likely to churn**, which became one of the key behavioral signals used in the churn prediction model.
- Detected and resolved **8–10% data quality issues**, including missing recharge dates, duplicate subscriber IDs, inconsistent plan codes, abnormal usage values, and mismatched customer records by coordinating with data engineering teams.
- Cleaned, transformed, and standardized large-scale subscriber datasets using **Pandas, NumPy, SQL, and Excel**, removing nearly **120K duplicate records** and improving the reliability of downstream segmentation and churn modeling.
- Engineered **25+ customer-level behavioral features**, including rolling 30/60/90-day usage averages, RFM metrics, call drop frequency, customer service contact rate, plan downgrade history, days since last recharge, recharge frequency, and contract expiry indicators.
- Built customer segmentation models using **K-Means Clustering** to group **4.2M active subscribers** into **6 behavior-based segments**, including high-value loyalists, at-risk heavy users, price-sensitive churners, dormant users, new activations, and stable mid-tier customers.
- Applied the **Elbow Method and Silhouette Score** to determine optimal cluster count and partnered with marketing stakeholders to profile each segment for targeted retention campaigns and customer engagement strategies.
- Developed churn prediction models using **Logistic Regression as a baseline** and **Random Forest as the final model** to capture non-linear customer behavior patterns and handle class imbalance where churn represented nearly **8%** of the subscriber base.
- Achieved model performance of **83% accuracy, 0.81 ROC-AUC, 74% precision, and 68% recall**, exceeding the client's production-readiness threshold of **0.78 ROC-AUC** for churn prediction.
- Generated monthly churn probability scores for **4.2M active subscribers**, enabling the retention team to prioritize high-risk customers and focus outreach on the top **180K at-risk subscribers** each quarter.
- Compared model results against legacy rule-based churn flags and identified approximately **22% additional at-risk subscribers** that were previously missed, improving campaign targeting and retention planning.
- Back-tested the churn model against **2 prior quarters of historical data** and collaborated with QA teams to resolve **3 critical data mapping issues** before stakeholder review and final project sign-off.
- Built Tableau dashboards and Excel Pivot reports to visualize **segment distribution, churn risk heatmaps, monthly segment migration, regional churn trends, plan-level risk patterns, and top churn drivers** for product and marketing teams.
- Helped drive a **15% reduction in quarterly churn** for the targeted subscriber group by enabling risk-based retention outreach, improving customer segmentation, and supporting campaign personalization.
- Automated monthly scoring and reporting workflows using **Python, SQL, and Excel-based reporting**, saving the client team approximately **2–3 days of manual effort per month** and improving reporting consistency.

Certifications

- AWS Certified Solutions Architect – Professional
- Google Cloud Professional Machine Learning Engineer

EDUCATION:

Gayatri Vidya Parishad College of Engineering
bachelor's in computer science and engineering

Aug 2011 – May 2015